CASE STUDY 07

Retail Billing with Multiple Discount Rules

1. PROBLEM STATEMENT

Prepare an itemized retail bill for multiple products. Apply category-specific tax, a discount based on the bill total, and a
maximum discount limit.

2. OBJECTIVES

• Calculate each product's amount from price and quantity.

• Apply category-based tax.

• Apply a threshold discount without exceeding the maximum concession.

3. ASSUMPTIONS

• Electronics tax is 18%; books tax is 5%; other items tax is 12%.

• A 10% discount applies when the subtotal is at least ■2,000.

• The discount cannot exceed ■500.

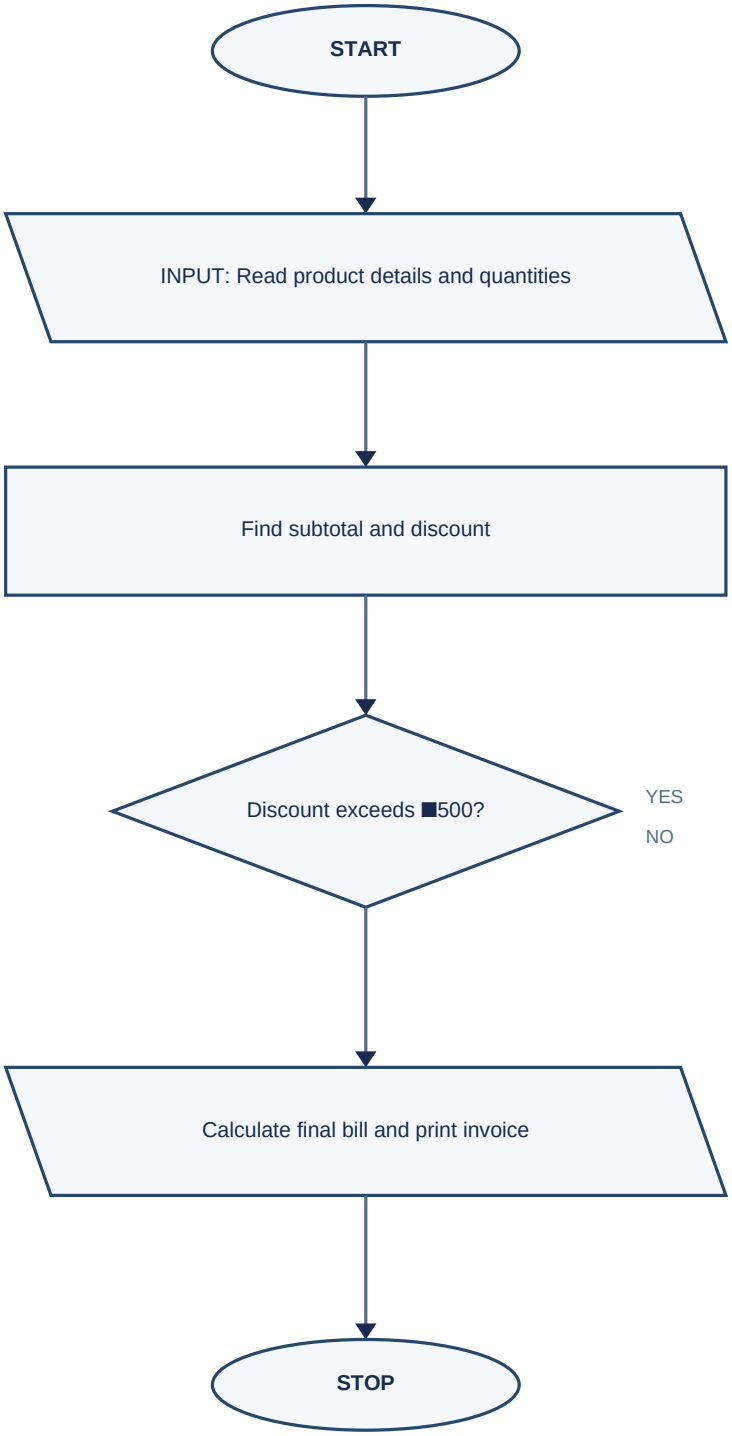
• Tax is calculated on the subtotal before discount for this exercise.

4. ALGORITHM

1. Store each product's name, category, price and quantity.
2. Calculate each item amount and add it to the subtotal.
3. Select the tax rate by category and calculate the tax total.
4. Apply a 10% discount when subtotal is at least ■2,000, limiting it to ■500.
5. Calculate and display the final bill with item-wise details.

5. FLOWCHART

Oval: START/STOP | Parallelogram: input/output | Rectangle: process/calculation | Diamond: decision



6. PSEUDOCODE

```
BEGIN
SET subtotal = 0, tax_total = 0
FOR each product
amount = price * quantity
ADD amount to subtotal
SELECT tax rate by product category
CALCULATE tax and add to tax_total
IF subtotal >= 2000: discount = 10% of subtotal
ELSE: discount = 0
IF discount > 500: discount = 500
final_bill = subtotal + tax_total - discount
DISPLAY itemized invoice and totals
END
```

7. PYTHON SOURCE CODE

```
products = [
["Keyboard", "electronics", 1200, 1],
["Python Book", "books", 500, 2],
["Notebook", "other", 100, 2]
]
subtotal = 0
tax_total = 0
print("ITEMIZED INVOICE")
for item in products:
name, category, price, quantity = item
amount = price * quantity
subtotal = subtotal + amount
if category == "electronics":
rate = 0.18
elif category == "books":
rate = 0.05
else:
rate = 0.12
tax = amount * rate
tax_total = tax_total + tax
print(name, "Qty:", quantity, "Amount:", amount,
"Tax:", round(tax, 2))
if subtotal >= 2000:
discount = subtotal * 0.10
else:
discount = 0
if discount > 500:
discount = 500
final_bill = subtotal + tax_total - discount
print("Subtotal: Rs.", subtotal)
print("Tax: Rs.", round(tax_total, 2))
print("Discount: Rs.", round(discount, 2))
print("Final Bill: Rs.", round(final_bill, 2))
```

8. TEST CASES & EXPECTED RESULTS

TEST INPUT / CONDITION REASONING EXPECTED RESULT

Keyboard ■1200 × 1 18% tax Tax ■216

Books ■500 × 2 5% tax Tax ■50

Subtotal ■2400 Subtotal ■2,400 + tax ■290 – discount ■240 Discount ■240; final bill ■2,450

9. SAMPLE OUTPUT

```
ITEMIZED INVOICE
Keyboard Qty: 1 Amount: 1200 Tax: 216.0
Python Book Qty: 2 Amount: 1000 Tax: 50.0
Notebook Qty: 2 Amount: 200 Tax: 24.0
Subtotal: Rs. 2400
Tax: Rs. 290.0
Discount: Rs. 240.0
Final Bill: Rs. 2450.0
```

10. CONCLUSION

The billing system calculates item totals, category taxes, and capped discounts to generate an itemized invoice.

SYLLABUS CONCEPTS Variables • Data Types • Operators • Conditions • Loops • Lists • Nested Lists • Tuples • String Operations